Brute Force Attack Detection – Splunk SIEM

i. Project Overview

This project demonstrates detection of brute-force login attempts using **Splunk SIEM** by monitoring Windows Security Event Logs.

The lab environment was configured to ingest Windows authentication logs into Splunk and analyze failed login attempts (Event ID **4625**) to identify potential brute-force activity. A real-time alert was created to trigger when multiple failed login attempts occur within a short time window.

This project highlights practical skills in **SIEM configuration, security log analysis, and automated threat detection**.

ii. Tools and Technologies

- Splunk SIEM
- Windows Event Logs
- Windows Security Event ID 4625 (Failed Login)
- PowerShell (attack simulation)
- Virtual Lab Environment

iii. Detection Logic

On Windows systems, every authentication attempt generates a security event log.

Relevant Event IDs:

- **4625** – Failed login attempt
- **4624** – Successful login attempt

Repeated **Event ID 4625 events** within a short time window may indicate a brute-force attack attempting to guess valid credentials.

The detection rule identifies accounts experiencing **multiple failed authentication attempts within five minutes**.

iv. Implementation

a. Log Ingestion

Windows Security Event Logs were forwarded to Splunk to enable centralized monitoring and analysis of authentication activity.

b. Splunk Detection Query

The following query was used to identify failed login attempts:

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4625
```

This query extracts failed authentication events from Windows Security logs.

c. Alert Configuration

A **real-time alert** was configured in Splunk to trigger when:

- Five or more failed login attempts occur
- Within a five-minute time window

This allows security teams to quickly identify potential brute-force attacks.

d. Attack Simulation

To validate the detection rule, simulated failed login attempts were generated using PowerShell:

```
runas /user:fakeuser cmd
```

This produced multiple failed authentication events, which were successfully ingested and detected by Splunk.

v. Key Outcomes

- Implemented centralized log monitoring using Splunk SIEM
- Developed a detection rule for identifying brute-force login attempts
- Configured real-time alerts for rapid incident awareness
- Validated detection accuracy through simulated attack activity

vi. Potential Enhancements

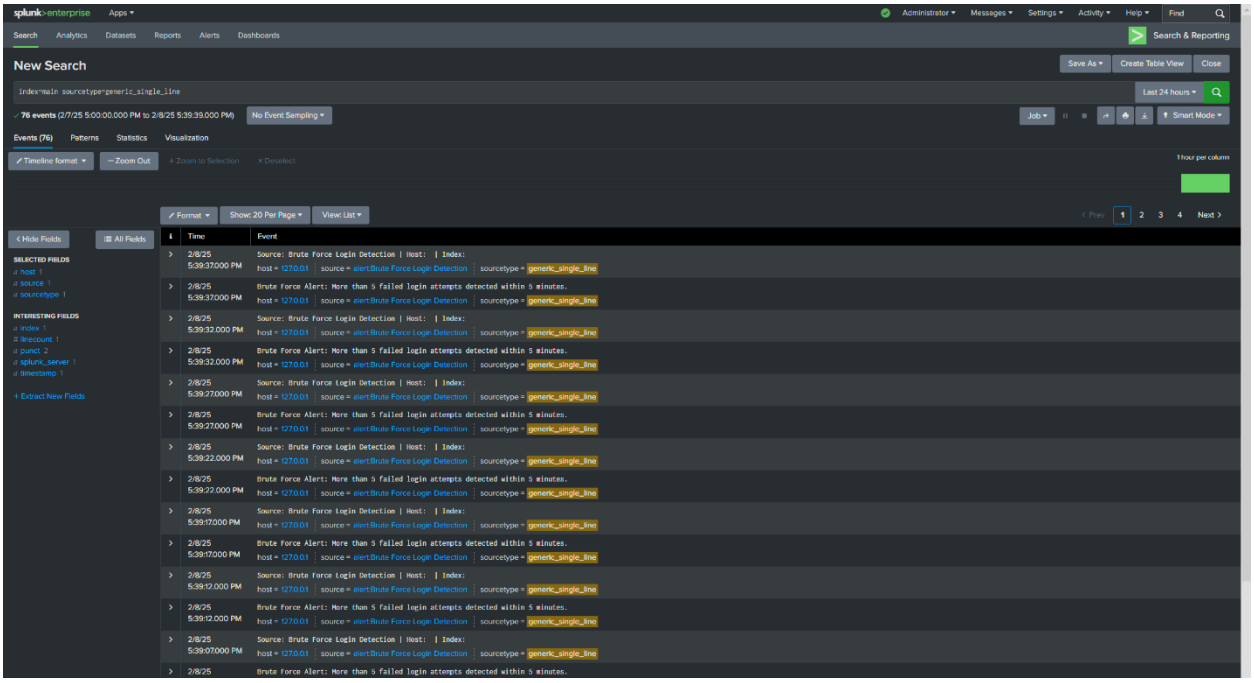
Future improvements could include:

- Detecting password spraying attacks across multiple accounts
- Correlating source IP addresses for attacker identification
- Creating Splunk dashboards to visualize authentication trends
- Implementing automated response workflows

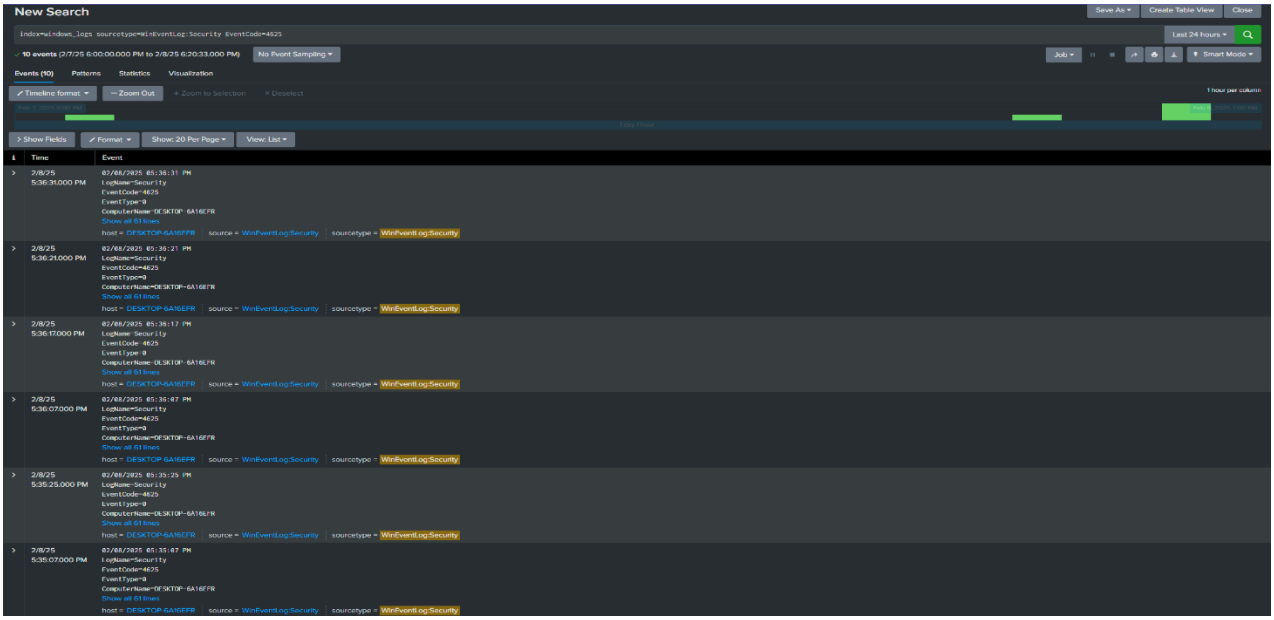
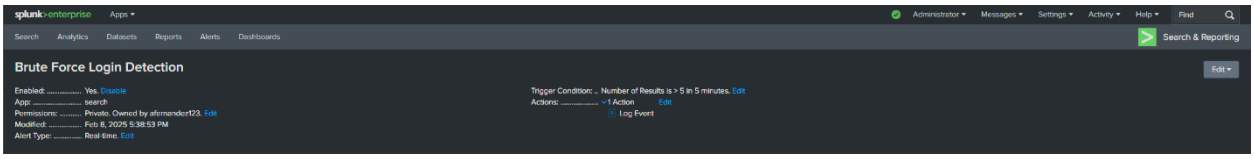
vii. Conclusion

This project demonstrates the ability to configure a **SIEM-based detection workflow** for identifying brute-force login activity. By integrating Windows event logs into Splunk and creating automated alerts, the system provides a practical foundation for **security monitoring and incident detection in a SOC environment**.

Splunk Query Detecting Brute Force Attacks



Windows Event Viewer – Failed Logins (Event ID 4625)



Splunk Alert Configuration – Brute Force Detection